

Session 32 Generalised muscle weakness: myasthenia gravis; myopathy and myositis; neuropathy; neurophysiology II

- ☐ To be able to evaluate a floppy baby clinically
- ☐ To differentiate floppiness with or without weakness in order to arrive at the underlying possible aetiology
- ☐ To understand the simple classification of neuromuscular diseases, esp the common types
- ☐ To recall the neuroanatomy of peripheral nerves, neuromuscular junction and muscle
- ☐ Neuropathy –
 - To understand different entities of neuropathy
 - To learn principles of neurophysiology tests in neuropathy
- ☐ Neuromuscular junction disorders –
 - To learn the classification, pathophysiology, pathogenesis, clinical features, investigations and differential diagnosis of Myasthenia gravis
- ☐ Myopathy –
 - To learn entities of muscle disorders
 - To learn principles of neurophysiology tests in myopathy

Generalized Weakness

- ☐ **Polyneuropathies**
- ☐ **Myopathies**
- ☐ **Neuromuscular junction disorders**

1. Polyneuropathies

1. Motor problems

- Weakness
- Wasting
- Fasciculations
- Hyporeflexia
- Bulbar, resp

2. Sensory problems

- Negative
- Positive

1. Polyneuropathies

Cardiovascular

Tachycardia, exercise intolerance
Cardiac denervation, painless myocardial infarction
Orthostatic hypotension
Heat intolerance
Alterations in skin blood flow

Gastrointestinal

Esophageal dysfunction
Gastroparesis diabeticorum
Diarrhea
Constipation
Fecal incontinence

Genitourinary

Erectile dysfunction
Retrograde ejaculation
Cystopathy
Neurogenic bladder

1. Polyneuropathies

□ Diversed aetiologies (> 100)

- Inflammatory demyelinating neuropathy
Guillain-Barre syndrome, CIDP
- Infectious and granulomatous neuropathy
Leprosy, sarcoid, HIV-related, Lyme disease
- Associated with systemic disease
Diabetes, chronic renal failure, vitamin deficiency, paraproteinaemias and other dysimmune neuropathies, connective tissue disease, critical illness polyneuropathy
- Ischaemic neuropathy
Peripheral vascular disease, vasculitis
- Metabolic neuropathy
- Hereditary neuropathy
- Toxins Pharmaceuticals, environmental toxins

1. Polyneuropathies

- **Time course**

Acute / subacute / chronic / relapsing

- **Distribution and pattern of deficits**

Generalized / focal

- **Type of functional fibres involved**

Motor / sensory / autonomic / mixed.

- **Primarily axonal, demyelinating, or mixed**

1. Polyneuropathies

□ Investigations

Electrophysiological studies

Look for systemic diseases

CSF

Nerve biopsy (seldom indicated)

Genetic tests

2. Myopathies

□ **Clinical features**

- Proximal weakness
- Waddling gait
- Bulbar, respiratory
- Tenderness
- Sensory, sphincter

2. Myopathies

Inherited

Muscular dystrophies

- Dystrophinopathies
- Facioscapulohumeral dystrophy
- Limb-girdle dystrophies

Metabolic

- Mitochondropathies
- Glycogenoses
- Lipid myopathies

Congenital myopathies

Channelopathies

- Myotonic disorders
- Periodic paralysis
- Malignant hyperthermia

Acquired

Inflammatory myopathies

- Polymyositis
- Dermatomyositis
- Inclusion body myositis
- Infection-related

Complicating systemic diseases

- Connective tissue diseases
- Endocrinopathies
- Sarcoidosis
- Critical illness myopathy

Toxic myopathies

Rhabdomyolysis

2. Myopathies

□ Investigations

Electrophysiological studies

Muscle enzymes (e.g., creatine kinase)

Look for systemic diseases

Muscle biopsy

Genetic tests

3. NMJ Disorders

- **Pre-synaptic**

- **Post-synaptic**

- Fluctuating weakness, fatigability
- Extra-ocular muscle manifestation
- Bulbar, respiratory
- Sensory, sphincter

Generalized weakness

- ☐ **Clinically important polyneuropathies**
- ☐ **Muscular dystrophies**
- ☐ **Inflammatory myopathies**
- ☐ **Myasthenia gravis**
- ☐ **Floppy infant syndrome**

- ☐ **Clinical neurophysiology**

Clinically important PN

- **Time course**

Acute / subacute / chronic / relapsing

- **Distribution and pattern of deficits**

Generalized / focal

- **Type of functional fibres involved**

Motor / sensory / autonomic / mixed.

- **Primarily axonal, demyelinating, or mixed**

Guillain-Barré syndrome

- “Rapidly progressive flaccid paralysis, areflexia and raised CSF protein without increase in cells”
- Incidence 1-2 per 100,000 per year
- Immune-mediated, inflammatory PN
- Antecedent viral/bacterial infection (URTI, GE) in 2/3 cases
- Subacute onset over a few days
- Nadir within 4 weeks

Guillain-Barré syndrome

- Ascending weakness with facial involvement
 - Respiratory failure 30%
 - Hypo- or areflexia
 - Sensory symptoms
 - Muscle ache
 - Autonomic dysfunction (> 2/3 patients)
- Investigations:
 - Raised CSF protein without pleocytosis
 - Electrophysiology

Guillain-Barré syndrome

□ **General supportive measures**

Monitor FVC, ABG, BP, ECG, bulbar function

Mechanical ventilation for respiratory failure

General care for immobility, nutritional support

Treat complications

□ **Specific immunotherapy**

Plasma exchange or high dose IV immunoglobulins

□ Mortality: 2 to 12%

20% permanent disabilities, 10% severe

Hereditary neuropathies

Charcot-Marie-Tooth disease

- Hereditary condition(s)
- Distal wasting
- Foot drop, pes cavus
- Slowly progressive

Hereditary neuropathies

□ Hereditary motor-sensory neuropathies

- Demyelinating or hypomyelination
- Axonal degenerative
- Autosomal dominant, recessive, X-linked
- *PMP22, MPZ, MFN2, GJB1 ...*

□ Hereditary motor neuropathies

- Spinomuscular atrophies
- *SMN1* (Spinraza), *UBA1, DYNC1H1, VAPB ...*

Charcot-Marie-tooth disease

HMSN-I (demyelinating PN) – AD, AR, X-linked

HMSN-II (axonal PN) – AD, AR, X-linked

HMSN-III (DSD)

HMSN-IV (Refsum disease)

HMSN-V (axonal PN with pyramidal signs)

HMSN-VI (with optic atrophy, HMSN-IIa)

HMSN-VII (with retinitis pigmentosa)

Spinal muscular atrophy

	SMA type	Onset	Milestones achieved	Evolution / natural history	Typical SMN2 copy number
60 %	1 A (also referred as type 0)	Prenatal	None	Death in weeks, contractures, cardiopathy	1
	1B	< 3M	Poor or absent head control	Feeding and respiratory problems, linear decline. Death by second or third year of life	2
	1C	> 3M	Cephalic control	Feeding and respiratory problems. Plateau in first two years	3
20-30 %	2	> 6M	Able to sit unaided	Scoliosis. Survival to adolescence/adulthood Weaker cases may lose sitting capability (2a) and stronger cases may stand with support (2b)	3
10-20 %	3a	Between 18 and 36 months	Walking unaided	Scoliosis Early loss of ambulation Normal lifespan	3
	3b	> 3 years	Walking unaided	Later loss of ambulation Normal lifespan	3-4
	4	Second or third decade of life	Walking unaided	Ambulant until late in life Normal lifespan	4

Motor neuron disease

- Idiopathic motor neuropathy
 - Progressive degeneration of α motor neuron:
 1. Anterior horn cells
 2. Corticospinal tract
 - ALS/PLS/PMA/PBP/Pseudobulbar palsy
 - Sensory involvement
 - Incidence: 1 per 100,000 per yr
 - Peak 6th decade, M:F = 3:2
 - Familial ~ 10% (*SOD1*, *C9orf72*, *TARDBP*, *FUS* ...)
- “Amyotrophic ... lateral sclerosis”.

Motor neuron disease

☐ **Treatment**

- No specific treatment
- Die within 3 years
- General nursing care
- Nutritional support
- ☐ - Riluzole, Endaravone (?)
- Euthanasia vs
long-term ventilation

Neurotoxic agents

- Alcohol
- Solvents and other industrial chemicals
- Heavy metals

- Drugs: Anti-neoplastic agents
 Anti-microbials (Tb, HIV)
 Others Colchicine, thalidomide,
 pyridoxine abuse

Muscular dystrophies

□ **Old definition**

1. Inherited disease
2. All symptoms due to muscle weakness
3. Progressive
4. No histopathological abnormalities other than degeneration and regeneration

□ **Clinical classification**

Duchenne (DMD), Becker (BMD)
Facioscapulohumeral (FSHD)
Limb-girdle (LGMD)
Myotonic dystrophy (MyoD)

X-linked muscular dystrophies

□ **Duchenne muscular dystrophy**

Mild delay in motor milestones

Progressive weakness from 3 - 5

Wheelchair-bound by 9 - 12

Respiratory failure by late teens

Cardiomyopathy, mild mental impairment

No or few dystrophin-positive fibres

□ **Becker muscular dystrophy**

Continue to ambulate beyond 15

Death between 40 - 60

Dystrophin stain - partial deficiency

Muscular dystrophies

□ Classification:

1. Phenotypic

2. Immunohistochemical

Dystrophin-opathy

Sarcoglycan-opathy

Laminin, caveolin, calpain, dysferlin ...

3. Genetic

DMD, BMD

Deletion in *dystrophin* gene

FSHD

SMCHD1, DUX4

LGMD

1A	<i>TTID</i>	1B	<i>LMNA</i>	1C	<i>CAV3</i>	ID	<i>DNAJB6</i>
1E	<i>DES</i>	1F	<i>TNPO3</i>	1G	<i>HNRPDL</i>		
2A	<i>CAPN3</i>	2B	<i>DYSF</i>	2C	<i>SGCG</i>	2D	<i>SGCA</i>
2E	<i>SGCB</i>	2F	<i>SGCD</i>	2G	<i>TCAP</i>	2H	<i>TRIM32</i>
2I	<i>FKRP</i>	2J	<i>TTN</i>	2K	<i>POMT1</i>	2L	<i>ANO5</i>
2M	<i>FKTN</i>	2N	<i>POMT2</i>	2O	<i>POMGNT1</i>	2Q	<i>PLEC1</i>

Inflammatory myopathies

Inflammatory infiltration (or necrosis) of skeletal muscle,
Limb-girdle pattern of weakness, occasionally myalgia,
Can be assoc with extra-muscular manifestations
Annual incidence: 5 /100,000

- 1. Polymyositis**
- 2. Dermatomyositis**
- 3. Overlap myositis**
- 4. Inclusion body myositis**
- 5. Necrotising autoimmune myopathy**

IDIOPATHIC INFLAMMATORY MYOPATHIES

Inflammatory infiltration (or necrosis) of skeletal muscle,
Limb-girdle pattern of weakness, occasionally myalgia,
Can be assoc with extra-muscular manifestations
Annual incidence: 5 /100,000

Bohan and Peter (1975)

Polymyositis, Dermatomyositis,
DM (PM) assoc with neoplasia, Childhood DM (PM),
PM/DM overlap with collagen-vascular disease

Dalakas (1991)

sIBM, amyopathic DM

IDIOPATHIC INFLAMMATORY MYOPATHIES

Troyanov (2005)

PM, DM, OM, CAM, sIBM, non-inflammatory myopathies

ENMC (2003)

Definite PM, probable PM,

Definite DM, probable DM, amyopathic DM, probable amyopathic DM,
sIBM, NAM, non-specific myositis

IMCCP (2017?)

International Myositis Classification Criteria Project

AUTOIMMUNE MYOPATHIES (AIM)

Antibody	Frequency in IIM, %	Clinical Significance and Associations
Myositis-specific autoantibodies		
Anti-Jo-1 (histidyl-tRNA synthetase)	15–20	Antisynthetase syndrome, including juvenile antisynthetase syndrome; sometimes clinically amyopathic
Anti-PL-7 (threonyl-)	5–10	
Anti-PL-12 (alanyl-)	< 5	
Anti-EJ (glycyl-)	5–10	
Anti-OJ (isoleucyl-)	< 5	
Anti-KS (asparaginy-)	< 5	
Anti-Zo (phenylalanyl-)	< 1	
Anti-Ha-YRS (tyrosyl-)	< 1	Immune-mediated necrotizing myopathy (generally severe and rapidly evolving with frequent myocardial involvement); rarely in children
Anti-SRP	5–10	
Anti-200/100 (HMGCR)	40% of necrotizing myopathy ¹⁰⁷	
Anti-Mi-2	5–30	Immune-mediated necrotizing myopathy, frequently associated with prior statin use
Anti-CADM-140 (MDA5)	50% of CADM	Classical DM, often of sudden onset with erythroderma and shawl sign, generally without ILD or malignancy and good prognosis; occasional juvenile DM
Anti-SAE	5%	DM, sometimes juvenile, with mild or absent muscle inflammation (CADM) and increased risk of ILD that can be rapidly progressive
Anti-p155(p140) (TIF1- $\alpha/\beta/\gamma$)	15–25% of adult DM, 40–75% of cancer-associated DM, 30% of juvenile DM	Adult DM
Anti-MJ (NXP-2)	< 5; 25% of juvenile DM	DM, especially cancer-associated (Ca in 45–75% of positive anti-TIF1- $\alpha/\beta/\gamma$ patients). Common in juvenile DM (without malignancy); rare in PM
Anti-PMS1	7.5	Juvenile DM exclusively; frequently severe cases with calcinosis
		Rare DM and PM
Myositis-associated autoantibodies		
Anti-U1RNP	10	OM, MCTD
Anti-Ku	20–30	PM-SSc overlap (Japanese)
Anti-PM-Scl	8–10	PM-SSc overlap (whites)
Sporadic inclusion body myositis (s-IBM)-associated autoantibodies		
Anti-43-kDa muscle autoantigen	50% of s-IBM ²² (needs further confirmation)	s-IBM

CADM: clinically amyopathic dermatomyositis; MCTD: mixed connective tissue disease; HMGCR: 3-hydroxy-3-methylglutaryl-coenzyme A reductase; ILD: interstitial lung disease; MDA5: melanoma-differentiation-associated gene 5; OM: overlap myositis; NXP-2: nuclear matrix protein 2; SAE: small ubiquitin-like modifier activating enzyme; SRP: signal recognition particle; SSc: systemic sclerosis; TIF1- $\alpha/\beta/\gamma$: transcriptional intermediary factors 1- α , β , and/or γ .

Lazarou and Guerne, J Rheumatol 2013

AUTOIMMUNE MYOPATHIES (AIM)

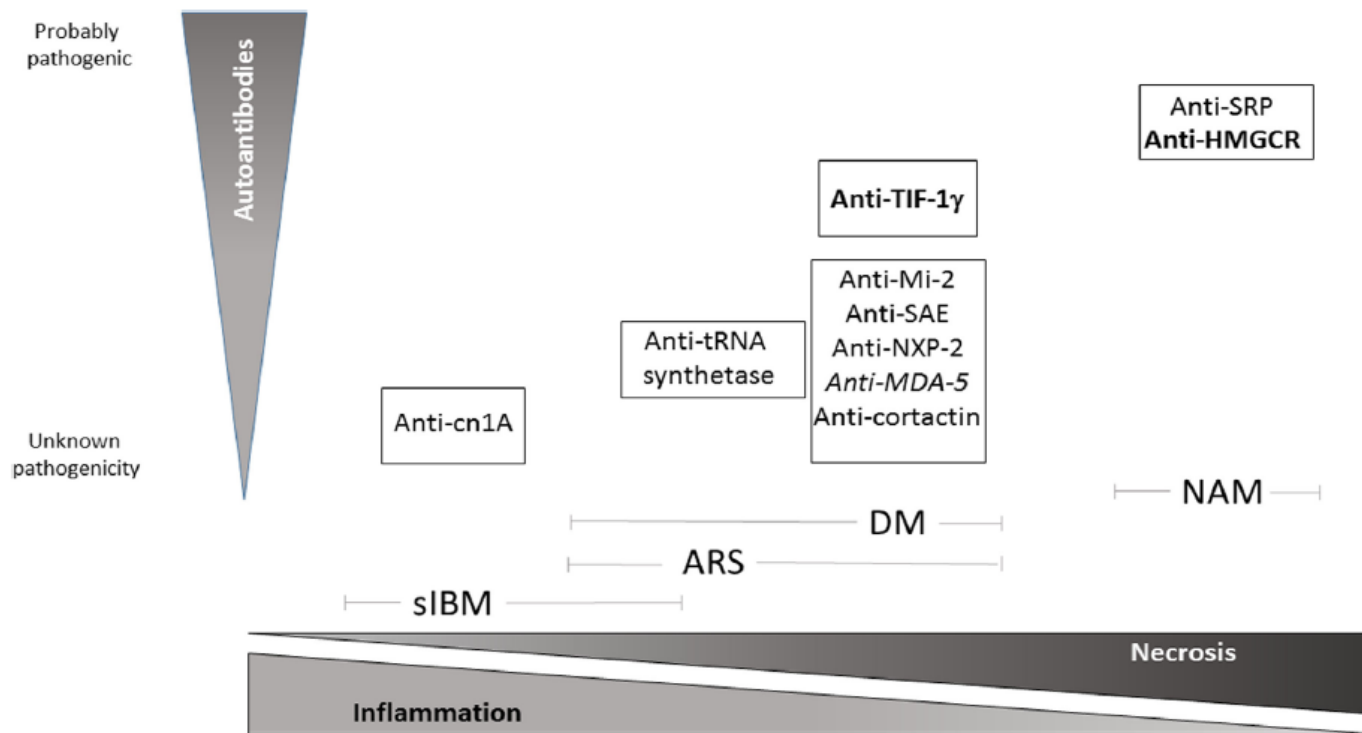


FIGURE 1 | Schematic view of pathological pattern of inflammatory myopathies and proven degree of pathogenicity of specific autoantibodies relative to each IM subtypes. Bold: high risk of neoplasia, Italics: amyopathic DM. NAM, necrotizing autoimmune myopathy; DM, dermatomyositis; ARS, anti-t-rna synthetase syndrome; sIBM, sporadic inclusion-body myositis.



INCLUSION BODY MYOSITIS

Commonest AIM $\geq$ age 50, M > F

Quadriceps, volar muscles, dorsiflexors $\rightarrow$ axial muscles

Slowly progressive, life expectancy not significantly affected

CK normal or up to 10-fold above ULN

Inflammatory infiltrates (CD8⁺T cells, macrophages) invading non-necrotic muscle fibres, rimmed vacuoles, congophilic inclusions (amyloidogenic proteins)

Anti-cytosolic 5'-nucleotidase 1A (cN1A)

Immuno-therapy not beneficial

DERMATOMYOSITIDES 1

Heterogenous group of inflammatory myopathies

F > M, particularly adult cases

Subacute or insidiously progressive proximal weakness, sometimes with myalgia

Skin manifestations (heliotrope rash, malar, V & shawl rash, Gottron papules, mechanic's hands, periungal erythema, gingival and oral involvement)

ILD, polyarthritis

Calcinosis and GI involvement in juvenile cases

Assoc with neoplasia in 6 – 45 % cases (breast, lung, pancreas, colon, cervix, NPC; 2 yrs before, 3 yrs after)



DERMATOMYOSITIDES 2

Complement-mediated microangiopathy: Perimysial and perivascular inflammation (macrophages, B-cells, CD4⁺ plasmacytoid dendritic cells), perifascicular atrophy, capillary depletion

Anti-TIF-1 γ	Neoplasia, dysphagia
Anti-NXP-2	Neoplasia in adults, calcinosis in children
Anti-Mi2	Rapid onset, but low risk neoplasia, ILD, good response to immunotherapy
Anti-MDA-5	Amyopathic DM, polyarthrititis, severe ILD
Anti-SAE	Amyopathic DM, in Asians assoc ILD

POLYMYOSITIS

“Symmetric proximal myositis and the absence of histopathological signs of other myopathies or typical rash of DM”

Endomysial mononuclear inflammatory infiltrates

Previously considered relatively prevalent

Actually a heterogeneous group of disorders; many PM cases in fact sIBM or OM

True idiopathic PM rare (or even a controversial entity)

Similarities with other AIM, hence “diagnosis of exclusion”

OVERLAP MYOSITIS

Troyanov classification: Myositis with overlap features (other than rash) and/or “overlap autoAb”

With other connective tissue diseases:

- Anti-U1RNP (SLE/MCTD-OM),

- Anti-PM/Scl, anti-Ku, anti-U3RNP (SSc-OM)

- Heterogeneous or non-specific muscle histology

Anti-synthetase syndrome:

- Acute myopathy, fever, ILD, polyarthritis, Raynaud's, mechanic's hand

- Anti-Jo1 (anti-tRNA-synthetase)

- Anti-PL-7, anti-PL-12 (amyopathic variant), anti-OJ, anti-KS, anti-EJ, anti-Zo, anti-Ha-YRS

MYOSITIS-AUTOANTIBODIES (60 – 80 % AIM)

Immunoprecipitation/immunodot, ELISA/ALBIA

PET SCAN (Malignancies in 20 % AIM)

ELECTROMYOGRAPHY

Increased insertional activity, fibrillation, CRD, spiky, polyphasic MUAP

MUSCLE MRI

Fat suppression or STIR: Signal changes in muscle and fascia from inflammation, edema, or muscle replacement by connective tissue

Guiding biopsy site (also EMG) and monitoring disease activity/treatment response

Myasthenia gravis

- ❑ Prevalence 5 per 100,000
- ❑ Antibody to ACh receptor
- ❑ Fluctuating weakness, fatigability
- ❑ Ocular $\pm$ generalized involvement
- ❑ Associated with
 1. Thymic pathology
 2. Other autoimmune diseases

Myasthenia gravis

□ Investigations

- Anti-AChR Ab
- Electrophysiology
Tensilon[®] test
- CT thorax
Thyroid function
± other autoantibodies

Myasthenia gravis

□ Treatment

Supportive measures

Specific treatment

- Anti-cholinesterase (Pyridostigmine)
- Thymectomy
- Immunotherapy

Pre-synaptic NMJ disorders

Lambert-Eaton myasthenic syndrome (LEMS)

- Ab to voltage-gated Ca channel
- Assoc with SCLC

Botulism

- Toxins from *Clostridium botulinum*
(food, wound, gut)
- Cytotoxic to nerve terminals

Floppy infant syndrome

Cerebral hypotonia

Chromosomal disorders

- Trisomy

- Partial chromosome deletion/duplication

- Prader-Willi syndrome

Static encephalopathy

- Cerebral malformation

- Perinatal distress

- Postnatal cerebral injury

- Idiopathic disorders

Single gene disorders

- Zellweger's syndrome

- Neonatal adrenoleukodystrophy

- Oculocerebral renal syndrome

- Acid maltase deficiency

- Carbohydrate-deficient glycoprotein syndrome

Spinal cord disorders

- Hyopxic-ischaemic myelopathy

- Trauma

- Congenital malformation

Floppy infant syndrome

Motor neuronopathies

- Spinal muscular atrophy
- Congenital cervical spinal muscular atrophy
- Infantile neuronal degeneration
- Neurogenic arthrogryposis
- Vaccine-associated poliomyelitis
- Incontinentia pigmenti

Polyneuropathies

- Congenital hypomyelinating polyneuropathy
- Dejerine-Sottas syndrome
- Idiopathic motor sensory polyneuropathies

Disorders of neuromuscular transmission

- Congenital defect of neuromuscular junction
- In utero passive transfer of maternal antijunction antibodies
- Autoimmune myasthenia gravis
- Infantile botulism

Myopathies

- Genetic myopathies
- Central core disease
- Congenital fibre-type disproportion myopathies
- X-linked myotubular myopathy

Clinical Neurophysiology

1. Nerve conduction studies

Assessment of large fibre conduction function

Analysis of neuronal signals generated by electrical activation

2. Needle electromyography (EMG)

Analysis of bioelectric activities of muscles

3. Special techniques for NMJ conduction functions

Clinical Neurophysiology

□ **Electrodiagnostic (EDx) studies**

- Objective assessment of PNS function
- Generate quantitative and reproducible information
- Non-invasive

Clinical Neurophysiology

Common diagnoses in electrodiagnostic medicine⁹

Myopathies	Idiopathic inflammatory myopathies (polymyositis, dermatomyositis), muscular dystrophies, other myopathies, myotonic disorders
Neuromuscular junction disorders	Myasthenia gravis, Lambert-Eaton myasthenic syndrome
Polyneuropathies	Demyelinating polyneuropathies, (acute and chronic inflammatory demyelinating polyradiculoneuropathy), hereditary motor sensory neuropathies, axonal polyneuropathies
Mononeuropathies	Carpal tunnel syndrome, ulnar neuropathy at the elbow, peroneal nerve palsy, other entrapment neuropathies, Bell's palsy
Multiple mononeuropathies	
Nerve trauma	
Plexopathies	
Radiculopathies	Cervical radiculopathy, lumbrosacral radiculopathy
Neuronopathies	Motor neuron disease, spinal muscular atrophy, poliomyelitis, dorsal root ganglionopathy

Clinical Neurophysiology

Indications for electrodiagnostic studies⁸

1. Identify normal and abnormal nerve, muscle, motor and sensory, and neuromuscular junction functioning.
2. Localize region(s) of abnormal function.
3. Define the type of abnormal function.
4. Determine the distribution of abnormalities.
5. Determine the severity of abnormalities.
6. Estimate the date of a specific nerve injury.
7. Estimate the duration of the disease.
8. Determine the progression of abnormalities or of recovery from abnormal function.
9. Aid in diagnosis and prognosis of disease.
10. Aid in selecting treatment options.
11. Aid in following response to treatment by providing objective evidence of change in neuromuscular function.
12. Localize correct locations for injection of intramuscular agents (e.g., botulinum toxin).

Clinical Neurophysiology

- To exclude/confirm peripheral neuropathy, myopathy, etc.
- To distinguish between neuropathy and myopathy
- Peripheral neuropathy clinically, to define aetiology
- Myopathy clinically, to define aetiology
- Cervical myelopathy.

Clinical Neurophysiology



- Diagnosing focal or multifocal neuropathies
- Demyelinating *vs* axonal degenerative neuropathies
- Assess severity of neuropathy
- Monitor progress of neuropathy/myopathy
- Monitor response to Rx
- Prognosis after nerve trauma + guide intervention
- EMG-guided BTx therapy.